Reducing steel waste: what we propose, what we need, what it costs

From: Banaov Technologies To: Cost Control team (for forwarding to management)

The short version

You told us you cannot currently say how much steel is wasted, how much of it could have been avoided, or what unused material is sitting in the yard. We suggest fixing that first, then using the answer to cut waste. We are not proposing to replace your nesting team or your Strumis system.

What we propose

A 20-week programme in three parts:

1. A waste report built from the data you already hold. It shows waste by project, by material and by cause, so management can see where steel is lost.
2. A fair test of nesting methods. We replay past jobs through your team's manual nesting, Strumis automatic nesting and our method, and compare the steel used. This settles whether manual nesting really saves more.
3. A suggestion tool for leftover steel. When new drawings are released, it points out offcuts and spare material that could be used before new steel is bought. Your nesters accept or reject every suggestion. Nothing changes in production without their approval.

When you will see results

- Week 4: the first waste report, so you have something concrete to show head office.
- Week 8: a decision point. We show the measured savings from the test. If they are too small, you can stop, keep the report and the test results, and pay only for the work done.
- Week 16: the suggestion tool in live use on two or three projects.
- Week 20: full handover and a measured results report.

What we need from you

- Read-only access to Strumis data by the end of week 2.
- A named contact in the nesting team, in stores, and in finance, each for about 2 hours a week.
- A decision on whether steel bought for one project can be used on another, and how the cost is moved between projects, by week 6.
- A short call every two weeks with someone from management.

What it costs

- Fixed price: $165,000, paid in five stages tied to the milestones above. By the week-8 decision point, $83,000 has been paid.
- Running cost after handover: about $31,000 a year (hosting and support).

What you could gain

Our estimate is a saving of about 560 tonnes of steel a year, roughly $308,000 a year, assuming a 0.7 percentage point improvement in overall yield. This is our estimate, not a promise. The week-8 test replaces it with a measured number. In a cautious case (0.4 points) the saving is about $176,000 a year and in a good case (1.0 point) about $440,000. At the base case, the cost is recovered about 12 months after we start.

We have assumed steel costs $800 a tonne and scrap sells for $250 a tonne; please correct us.

Other areas you mentioned

Yard layout, trailer loading, drawing revisions and equipment use are all worth looking at, but they depend on the same data and on other teams. We suggest doing them after this programme. If management wants a quick answer on trailer and equipment use, we can analyse your CarTrack and booking data in 2 to 3 weeks for an extra $12,000.

One question for the next call

Roughly what share of your steel, by weight, is plate and what share is sections? Your waste figures only add up if plate is a small share, and the answer changes where we focus.